

More on Step Response and Pendulum Modeling

EGR 345

GVSU

Outline

- some thoughts on lab
- review: MSD step response
- more pendulum modeling

Dialing in the right difficulty

- why can't I just give you step-by-step instructions?
 - what happens if I make it too easy?
- what happens if I make it too hard?

Blame former students

- trying to write detail instructions didn't go well for me...

You can always ask me questions

- it is ok to interrupt “off topic” conversations
- how do I know if you are struggling?
- I will never yell at you for asking a questions
 - worst case, I ask you questions and point you in a direction to get your own answers

Complaint from last year

- We were all struggling with lab while he was joking around and playing chess on his phone...
- if I think you are all working hard and no one is asking me any questions...
- I have never delayed to answer a question.

Lab is a work in progress

- rearranging the lectures to focus on MSD systems isn't too hard
- I don't have the perfect MSD experiments yet
 - but you can help with that. . .

Lab Plans

- investigate modeling the pendulum as an MSD system
 - initial condition response and tap input (last week)
 - voltage time domain inputs (pulse and step) (this week)
 - Bode system id (next week)
- pendulum vibration suppression
- MSD system design project

Lab Conclusion

- If you don't ask me questions, I won't know you are struggling.

Review

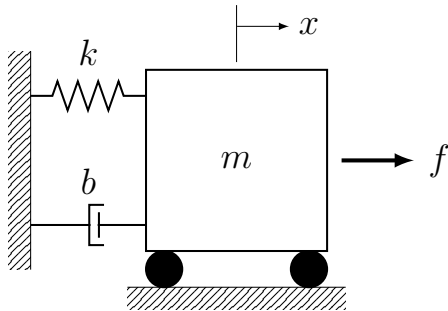
MSD Review

Find the unit step response for an MSD system using the parameters:

- $m = 2$
- $k = 75$
- $b = 1$

verify your answer with Python

assume zero conditings



$x(t)$

→ FBD

→ Newton's 2nd Law → EOM

→ Laplace

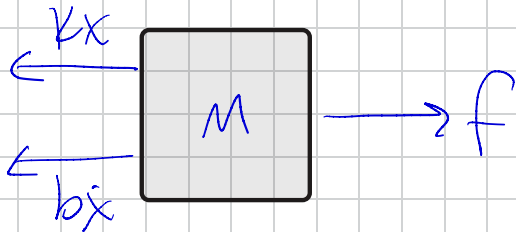
→ TF

→ $X(s) = G(s) \cdot U(s)$

→ PFE → $\mathcal{L}^{-1}$ → $x(t)$

$\swarrow \frac{1}{s}$

FBD



$$M\ddot{x} = f - b\dot{x} - kx$$

$$m\ddot{x} + b\dot{x} + kx = f \rightarrow (ms^2 + bs + k)x = f$$

$$G(s) = \frac{X(s)}{f(s)} = \frac{1}{ms^2 + b s + k} = \frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}}$$

$\uparrow$
 $ms^2 + b s + k$

$$G(s) = \frac{1/m}{s^2 + \frac{b}{m}s + \omega_n^2}$$

$$b_{cr} = 24.49$$

$$\xi = \frac{1}{24.49} \quad \checkmark$$

$$b_{cr} = 2\sqrt{km}$$

$\uparrow$
 $b = \xi b_{cr}$

$$\frac{\xi \cdot 2\sqrt{km}}{m} = 2\xi \sqrt{\frac{k}{m}} = 2\xi \omega_n$$

$$G(s) = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$Y(s) = \frac{1}{s} \cdot \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$0 < \zeta < 1 \rightarrow$ complex roots $\rightarrow$ rows 8+9

$$\frac{1}{s} \frac{L/M}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{C_1}{s} + \frac{C_2 s + C_3(s+a)}{(s+a)^2 + \omega^2}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + 2\zeta\omega_n s + a^2 + \omega^2$$

$$C_1 = \frac{1}{75}$$

$$C_3 = -C_1$$

$$a = \zeta\omega_n = \frac{1}{2H\pi} \sqrt{\frac{F_1}{2}} = 0.25$$

$$\omega = 6.12 \approx \omega_d$$

$$C_2 = -0.000544$$

$$2\zeta\omega_n = 2a$$

$$\left[\frac{Y_M}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{C_1}{s} + \frac{C_2 s + C_3(s+a)}{(s+a)^2 + \omega^2} \right] (s(s^2 + 2\zeta\omega_n s + \omega_n^2))$$

$$\frac{1}{n} = C_1(s^2 + 2\zeta\omega_n s + \omega_n^2) + (C_2 s + C_3 s(s+a))$$

$$\frac{1}{n} = \underline{C_1 s^2} + \underline{2\zeta\omega_n s C_1} + \underline{C_1 \omega_n^2} + \underline{C_2 s^2} + \underline{C_3 s^2} + \underline{C_3 s a}$$

$$s^0: \frac{1}{m} = C_1 \omega_n^2 = C_1 \frac{K}{m}$$

$$C_1 = \frac{1}{K} = \frac{1}{75} \quad \checkmark$$

C_1 checks out from steady-state
equilibrium

$$m\ddot{x} + b\dot{x} + kx = f$$

$x_{ss} = \frac{f_{ss}}{k}$

what happens
at steady-state?

$$X(s) = \frac{C_1}{s} + \frac{C_2 \omega + C_3 (s+a)}{(s+a)^2 + \omega^2}$$

$$x(t) = C_1 \cdot 1(t) + C_2 e^{-at} \sin(\omega t) + C_3 e^{-at} \cos(\omega t)$$

$$\left[x_{ss}(t) = C_1 = \frac{1}{k} \text{ (previous slide)} \right]$$

How to check your C_1 value

$$x(t) = C_1 \cdot 1(t) + C_2 e^{-at} \sin(\omega t) + C_3 e^{-at} \cos(\omega t)$$

$$x(0) = 0 = C_1 + C_3 \rightarrow C_3 = -C_1$$

$$1(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } t \geq 0 \end{cases}$$

$$S^2: 0 = C_1 S^2 + C_3 S^2 \rightarrow C_3 = -C_1$$

$$S^1: 0 = \cancel{2\omega_n C_1} + \cancel{C_2 \omega} + \overset{-C_1}{\underset{\uparrow}{C_3}} \omega$$

$$0 = 2\omega_n C_1 + C_2 \omega - \overset{\omega_n \uparrow}{C_1} \omega$$

$$0 = \cancel{2\omega_n C_1} - \cancel{\omega_n C_1} + C_2 \omega$$

$$0 = \omega_n C_1 + C_2 \omega \Rightarrow C_2 = \frac{-\omega_n C_1}{\omega}$$

$$C_2 = \frac{-\zeta \omega_n C_1}{\omega}$$

$$\omega = \omega_n \sqrt{1 - \zeta^2}$$

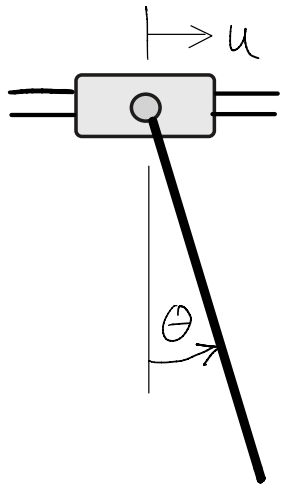
$$C_2 = \frac{-\zeta \omega_n C_1}{\omega_n \sqrt{1 - \zeta^2}} = \frac{-\zeta C_1}{\sqrt{1 - \zeta^2}} = \frac{1}{k} \cdot \frac{-s}{\sqrt{1 - s^2}}$$

Pendulum Modeling

Is a pendulum basically the same as a mass/spring/damper system?

- what did you learn in last week's lab?
 - initial condition response
 - impulse response from tap/whack
- what about voltage input to the motor on the cart?

Is a pendulum basically the same as a mass/spring/damper system?



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